MESTRADO EM ENGENHARIA INFORMÁTICA – Preparação para a Dissertação (PREPD)

2026/2027

Systematic Literature Review & State of the Art Module

From searching papers to identify research gaps

**“You have to start your Master’s dissertation tomorrow.
How would you find out what is already known?”**

Ana Maria Madureira

1

01 · PURPOSE

02

What is Systematic Literature Review (SLR)?

- **Literature review is a research activity intended to organize and summarize the research work already done in the area of interest**
- **Literature reviews are secondary studies – because they review original (primary) studies**

2

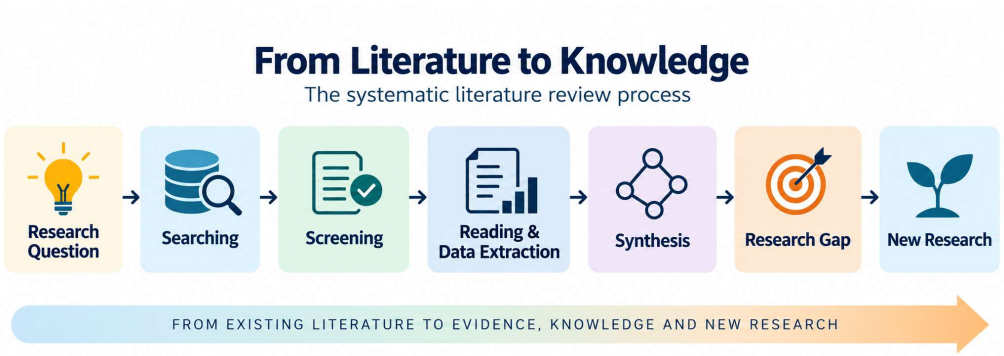
Literature Review Goals

- Provide a solid theoretical foundation for the proposed study in the dissertation course (related to “what is already known?”)
- To help the student understand the existing body of knowledge including where excess research exists (i.e. what is already know?) and where new research is needed (i.e. what is needed to be known?)
- Substantiate the presence of the research problem (related to “what is needed to be known?”)
- Frame the valid research methodologies, approach, goals, and research questions for the proposed study (dissertation)
- In certain cases, justify the proposed study as one that contributes something new to the body of knowledge

What is Systematic Literature Review (SLR)?

A systematic literature review (often referred to as a systematic review) is a means of identify, evaluate and interpret available research relevant to a particular research question, or topic area, or phenomenon of interest. Individual studies contributing to a systematic review are called primary studies; a systematic review is a form of secondary study.

From Literature to Knowledge



Reasons for Performing Systematic Literature Review?

- To summarize the existing evidence concerning a treatment or technology e.g. to summarize the empirical evidence of the benefits and limitations of a specific agile method.
- To identify any gaps in current research in order to suggest areas for further investigation.
- To provide a framework/background in order to appropriately position new research activities.

01 · PURPOSE

02

Why do we review the literature?

UNDERSTAND	COMPARE	EVALUATE	IDENTIFY GAPS	POSITION
What is known?	What approaches exist?	How strong is the evidence?	What remains unresolved?	Why is our work needed?

A literature review should explain not only what exists, but why new research is still needed.

02 · CONCEPTS

03

Literature Review ≠ State of the Art ≠ SLR

Literature Review	State of the Art	Systematic Literature Review
• Broad overview	• Focus on current frontier	• Explicit protocol
• May be narrative	• Critical & comparative	• Reproducible search
• Contextualises the topic	• Identifies limitations/gaps	• Transparent selection
• Flexible search	• Positions new work	• Systematic synthesis

A Master's dissertation can have a rigorous, systematised State of the Art without claiming to be a full SLR.

Support: Kitchenham & Charters (2007); Snyder (2019).

From Literature to Knowledge

The systematic literature review process



FROM EXISTING LITERATURE TO EVIDENCE, KNOWLEDGE AND NEW RESEARCH

Searching ≠ Screening ≠ Reading ≠ Synthesis

A literature review is not simply a search for papers: each stage has a different purpose.

SEARCHING

What exists?

Identify potentially relevant literature using a documented search strategy.

Result: candidate records.

SCREENING

What is relevant?

Apply explicit inclusion and exclusion criteria.

Result: eligible studies.

READING + EXTRACTION

What does the evidence say?

Critically read included studies and extract comparable information aligned with the RQs.

Result: structured evidence.

SYNTHESIS

What do we learn?

Integrate evidence to identify patterns, contradictions, limitations and research gaps.

Result: knowledge.

The value of a literature review is not in how many papers you find, but in how systematically you transform literature into evidence and knowledge.

02 · CONCEPTS

04

Three levels of literature review

Narrative Review — **Structured State of the Art** — **Systematic Literature Review**

Increasing methodological formalisation →

Search transparency	Selection protocol	Quality assessment	Reproducibility	Structured synthesis

Source: Snyder (2019); Petersen, Vakkalanka & Kuzniarz (2015).

03 · METHOD

05

The SLR Process



Traceability principle: every methodological decision should be connected to the Research Questions.

Source: Kitchenham & Charters (2007); Okoli (2015).

04 · RESEARCH QUESTIONS 06

Start with the Research Question

Too broad

How is Artificial Intelligence used in healthcare?

Better focused

How are explainable AI techniques being used to support clinical decision-making?

Operational / reviewable

Which XAI techniques have been applied to deep-learning-based clinical decision support systems, and how has their explainability been evaluated?

04 · RESEARCH QUESTIONS 07

Breaking the question into concepts

Concept 1 · XAI	Concept 2 · Decision Support	Concept 3 · Context
• explainable AI	• clinical decision support	• healthcare
• XAI	• CDSS	• clinical
• interpretability	• decision support system	• medical
• explainability	• decision-making	• patient care

(XAI OR "explainable AI" OR interpretability)
AND ("clinical decision support" OR CDSS) AND (healthcare OR clinical OR medical)

05 · SEARCH STRATEGY

08

Where should we search?

IEEE Xplore

Computing, electronics, engineering

ACM Digital Library

Computer science & computing

Scopus

Broad bibliographic index

Web of Science

Multidisciplinary citation index

ScienceDirect / SpringerLink

Publisher platforms

Google Scholar

Discovery & complementary searching

Database choice must be justified by disciplinary coverage — not by convenience.

Source: Kitchenham & Charters (2007); Wohlin (2014).

05 · SEARCH STRATEGY

09

Building Search Strings

GOOD: concept blocks + synonyms

("explainable artificial intelligence" OR "explainable AI" OR XAI)
AND ("clinical decision support" OR CDSS)
AND ("deep learning" OR "neural network*")

WEAK: underspecified

AI AND healthcare
AND explainability

AND = intersection · OR = synonyms/alternatives · "..." = phrase · * = truncation/wildcard

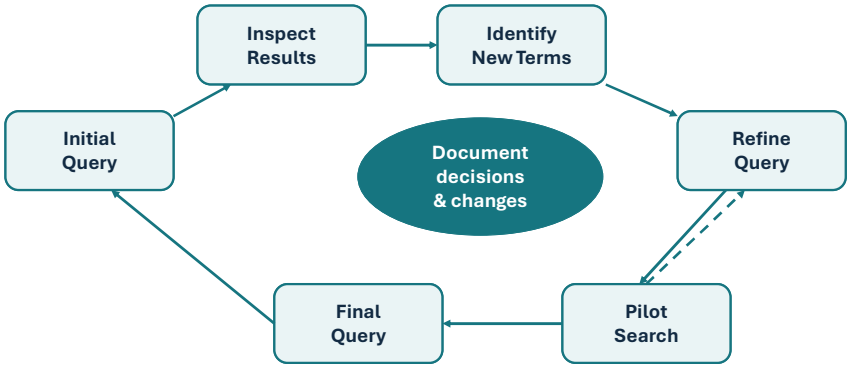
Class prompt: What relevant papers might the weak query miss?

Support: Kitchenham & Charters (2007).

05 · SEARCH STRATEGY

10

Search Strategy is iterative



Stop refining when the query is sufficiently **sensitive, specific and reproducible** to address the RQs.

Complement database search with ← Backward snowballing | Forward snowballing →

Source: Wohlin (2014), Guidelines for Snowballing in Systematic Literature Studies; Kitchenham & Charters (2007).

17

06 · SELECTION

11

Inclusion and Exclusion Criteria

INCLUSION	EXCLUSION
• Peer-reviewed publications	• Duplicates
• 2020–2026 (if justified)	• Editorials / abstracts only
• English language	• Outside target domain
• Target AI/ML method applied	• No empirical evaluation
• Empirical evaluation	• Outside research scope
• Full text accessible	• Insufficient methodological detail

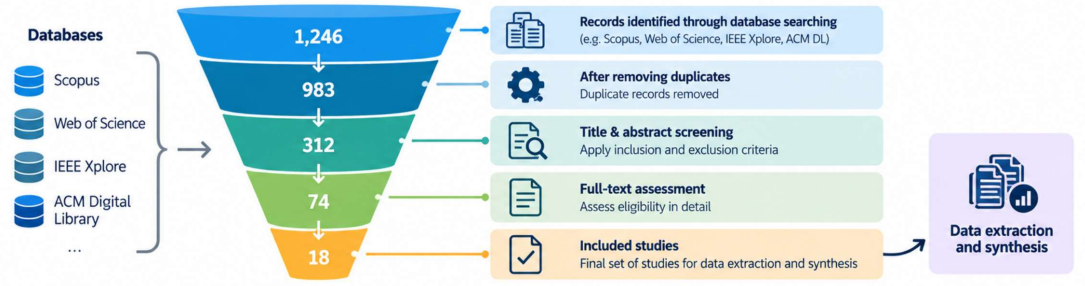
Criteria should be operational enough that another reviewer can apply them consistently.

Source: Kitchenham & Charters (2007); Snyder (2019).

18

The Funnel of a Systematic Literature Review

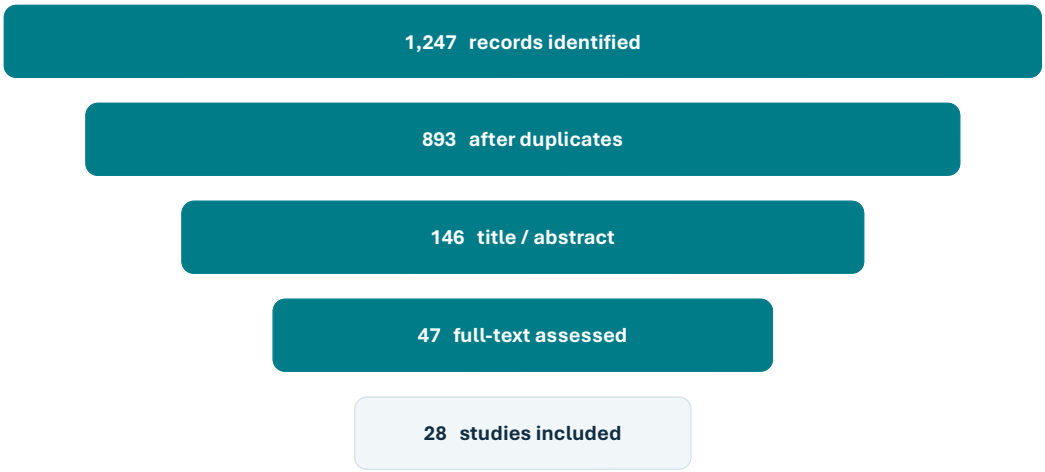
From many records to a focused set of studies



Criteria should be operational enough that another reviewer can apply them consistently.

Source: Kitchenham & Charters (2007); Snyder (2019).

Screening Process

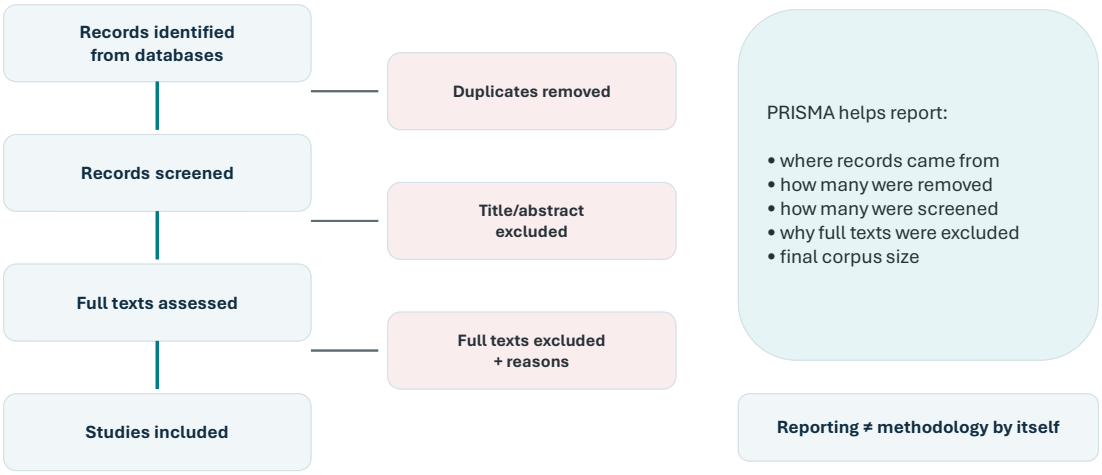


Source: Kitchenham & Charters (2007); Page et al. (2021).

06 · SELECTION

13

PRISMA: making selection transparent



PRISMA helps report:

- where records came from
- how many were removed
- how many were screened
- why full texts were excluded
- final corpus size

Reporting ≠ methodology by itself

Source: Page et al. (2021), PRISMA 2020.

07 · QUALITY ASSESSMENT

14

Are all selected papers equally useful?

Objectives clearly defined?	Yes 1 · Partial .5 · No 0
Method adequately described?	Yes 1 · Partial .5 · No 0
Dataset appropriate?	Yes 1 · Partial .5 · No 0
Metrics suitable?	Yes 1 · Partial .5 · No 0
Baselines/comparators appropriate?	Yes 1 · Partial .5 · No 0
Threats to validity discussed?	Yes 1 · Partial .5 · No 0
Conclusions supported by results?	Yes 1 · Partial .5 · No 0

Quality scoring should support judgement — not replace it.

Support: Kitchenham & Charters (2007); Snyder (2019).

08 · DATA EXTRACTION

15

Data Extraction: from papers to evidence

Study	Year	Problem	Method	Dataset	Metrics	Results	Limitations
S1	2024	Diagnosis	CNN	Private	F1	0.89	Small dataset
S2	2025	Triage	Transformer	Public	AUC	0.93	No ext. validation
S3	2026	Decision	Hybrid	Multi-site	F1/AUC	↑ robustness	Limited XAI

Extract what you need to answer the RQs — not everything that appears in the paper.

Define the extraction form before reading the final corpus in depth.

Source: Kitchenham & Charters (2007); Okoli (2015).

09 · CRITICAL SYNTHESIS

16

A State of the Art is NOT a sequence of summaries

DESCRIPTIVE LISTING

Author A proposed X.

Author B proposed Y.

Author C used Z.

Author D achieved 94%.

WHAT IS MISSING?

- Patterns across studies
- Comparable dimensions
- Reasons for differences
- Strengths and limitations
- Contradictions in evidence
- Unresolved research questions

Question to students: After reading the left-hand text, what have we actually learned?

From Description to Critical Synthesis

GROUP — COMPARE — EXPLAIN — CRITIQUE — IDENTIFY GAP

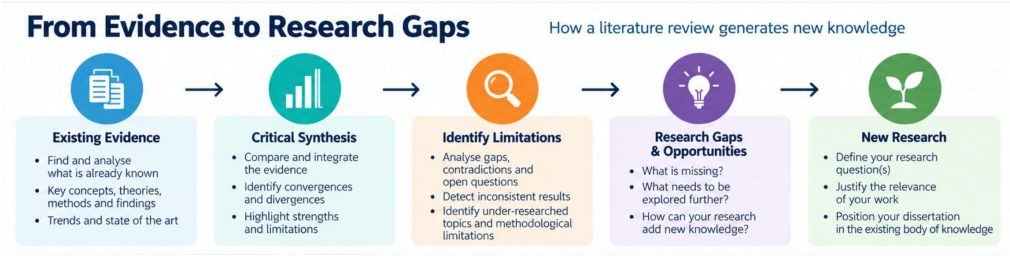
Existing approaches can be grouped into three main categories: CNN-based, transformer-based and hybrid architectures. Although transformer-based approaches report higher performance in several studies, direct comparison remains difficult because datasets and evaluation protocols differ substantially. Moreover, only a limited number of studies perform external validation...

Synthesis answers: What does the body of evidence collectively tell us?

Source: Webster & Watson (2002); Snyder (2019).

25

From Evidence to Research Gaps



A research gap should emerge from the critical synthesis of evidence; it should not simply be declared before the literature has been analysed.

26

10 · RESEARCH GAP

18

Finding the Research Gap

Knowledge	Methodological	Evaluation	Data
Insufficiently understood	Existing methods limited	Weak validation / metrics	Missing / unrepresentative data

Context	Technology	Integration	Trust / XAI
Not tested in target setting	Emerging approach unexplored	Solutions not integrated	Performance without trust

A gap is not "nobody used my favourite algorithm". It is an evidence-supported unresolved need.

Source: Snyder (2019); Webster & Watson (2002).

10 · RESEARCH GAP

19

From Evidence to Research Contribution

EVIDENCE	LIMITATION	GAP	RQ	CONTRIBUTION
High classification accuracy	Small controlled datasets	Robustness under distribution shift unclear	How robust is the model in realistic shifts?	Develop + evaluate robust approach

The contribution is strongest when the chain can be defended with evidence at every step.

Evidence → Limitation → Gap → Research Question → Contribution

Source: Snyder (2019); Okoli (2015).

Can AI do my Literature Review?

AI CAN SUPPORT

- Keyword / synonym generation
- Query refinement
- Preliminary categorisation
- Comparison frameworks
- Language improvement
- Brainstorming possible gaps

AI SHOULD NOT REPLACE

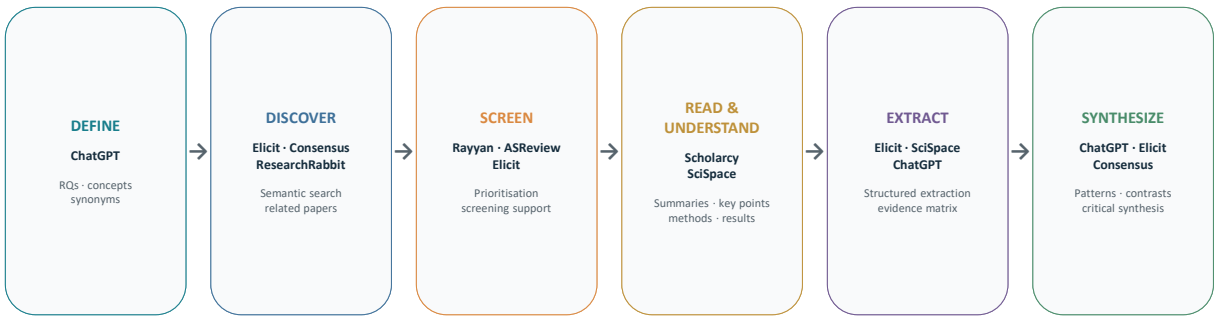
- Database searching evidence
- Reference verification
- Eligibility decisions
- Quality assessment judgement
- Interpretation of evidence
- Scientific responsibility

Use AI to augment the review process — not to outsource scientific judgement.

Source: Adel & Alani (2025) — risks of hallucination and unreliable synthesis with GenAI.

AI-Assisted Literature Review: Where Can AI Help?

AI can augment the workflow — but the protocol, verification and scientific judgement remain human responsibilities.



AI-generated summary ≠ Original evidence
Use summaries to support understanding — verify screening decisions, extracted facts and claims against the original paper.

HUMAN JUDGEMENT: TRACE · VERIFY · DOCUMENT · DECIDE
AI tools complement — they do not replace — bibliographic databases, the review protocol or researcher accountability.

11 · AI-ASSISTED REVIEWING

Acceptable vs. Unacceptable Uses of AI

A 5-minute critical exercise: decide whether each use preserves traceability, reproducibility and researcher judgement.

ACCEPTABLE — with verification

- ✓ Refine RQs and generate candidate synonyms
- ✓ Critique a Boolean search string
- ✓ Prioritise records for human screening
- ✓ Draft an extraction matrix from predefined fields
- ✓ Compare evidence already verified in the papers
- ✓ Improve wording of a synthesis written from validated evidence

UNACCEPTABLE / HIGH RISK

- ✗ Ask AI to 'find all relevant papers' and stop there
- ✗ Use references without checking the original source / DOI
- ✗ Let AI make final inclusion/exclusion decisions autonomously
- ✗ Accept extracted values without checking the paper
- ✗ Claim a research gap only because AI suggested it
- ✗ Generate the final review without documenting AI use

Decision test: Can I trace it? Can I verify it? Can I reproduce it? Who made the scientific decision?

31

12 · GROUP ACTIVITY

21

Practical Activity — Mini-SLR Protocol

Topic: Artificial Intelligence for detecting misinformation in social media

1. **Define 2 Research Questions**
2. Identify 3–4 concept groups + synonyms
3. Build 1 complete Boolean search string
4. Define 4 inclusion + 4 exclusion criteria
5. Design a data extraction table with 6 fields

20 MIN

Groups of 3–4

Output: 1-page mini-protocol

Critical question: What result would convince you that there is a genuine research gap?

32

12 · PEER REVIEW

22

Compare the Protocols

PEER REVIEW LENS

- ✓ RQ ↔ Search String
- ✓ Coverage of synonyms
- ✓ Scope: too broad / too narrow?
- ✓ Operational selection criteria
- ✓ Extraction ↔ RQs
- ✓ Potential sources of bias
- ✓ Reproducibility

QUESTIONS FOR DISCUSSION

- Would this query answer the RQs?
- What relevant terminology is missing?
- Which criterion may introduce bias?
- Can another group reproduce the process?
- What evidence would support the proposed gap?

13 · QUALITY CHECK

23

Common Mistakes + Master's Dissertation Checklist

COMMON MISTAKES

- Searching without RQs
- Using only Google Scholar
- Selecting "interesting" papers ad hoc
- Not documenting queries / dates
- Unjustified time window
- Citation count = quality
- One paragraph per paper
- Comparing incomparable datasets
- Claiming a gap without evidence
- Trusting AI-generated references

CAN I EXPLAIN...?

- What is already known?
- How was literature searched?
- Why were studies selected?
- What are the main approaches?
- How do they compare?
- What are their limitations?
- What remains unresolved?
- How does my work address the gap?

14 · SYNTHESIS

24

Take-Home Message

SEARCH — SELECT — EVALUATE — SYNTHESISE — GAP — CONTRIBUTE

A good literature review	maps knowledge.
A good State of the Art	critically evaluates the frontier.
A good SLR	makes the process transparent and reproducible.

The goal is not to collect papers. The goal is to build a defensible argument for research.

10 · REFERENCES

25

References & Further Reading

- Adel, A. & Alani, N. (2025). Can generative AI reliably synthesise literature? Exploring hallucination issues in ChatGPT. AI & Society, 40, 6799–6812.
- Kitchenham, B. & Charters, S. (2007). Guidelines for Performing Systematic Literature Reviews in Software Engineering. EBSE-2007-01.
- Okoli, C. (2015). A Guide to Conducting a Standalone Systematic Literature Review. Communications of the Association for Information Systems, 37. DOI: 10.17705/1CAIS.03743
- Page, M. J. et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ, 372:n71. DOI: 10.1136/bmj.n71
- Petersen, K., Vakkalanka, S. & Kuzniarz, L. (2015). Guidelines for conducting systematic mapping studies in software engineering: An update. Information and Software Technology, 64, 1–18.
- Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. Journal of Business Research, 104, 333–339. DOI: 10.1016/j.jbusres.2019.07.039
- Webster, J. & Watson, R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. MIS Quarterly, 26(2), xiii–xxiii.
- Wohlin, C. (2014). Guidelines for Snowballing in Systematic Literature Studies and a Replication in Software Engineering. EASE 2014. DOI: 10.1145/2601248.2601268

Recommended starting point for Computer Science / Software Engineering: Kitchenham & Charters + Wohlin + PRISMA 2020